Resources

STAT 109 — Summer Session 1, 2026 (online, async)

Resources

Everything you **read and practice** lives on this website. Everything you **submit for a grade** lives in Canvas. The course path is the recommended order through six modules (June 1 – July 2, 2026).

Start here

Resource	Link	When to use
Course path	Schedule	Every week — readings, labs, quizzes, project due pattern
Syllabus	Syllabus	Policies, grading, AI expectations
Assessments	Assessments	Weights, quiz types, project list, notebook rubric
Lectures index	Lectures	Module outcomes and course-path tables

Each module table on the course path includes a **Supplemental (optional)** column — textbook chapters and videos that match our Colab + `{tidyverse}` stack. None of those are required for the module project unless a row says otherwise.

Reference sheets (open notes on quizzes)

You will build and update these across Modules 1–5:

Sheet	Link	Purpose
Methods map	Methods	When to use which test or summary; study-design vocabulary
R reference	R reference	Code patterns, pipe, `ggplot2`, project checklists

Keep personal copies (Google Doc, Quarto, or notebook appendix) and update them after each module lab.

R and notebooks — Google Colab (primary)

This async course runs **R in Google Colab**. Lab and project pages link to notebooks on the public GitHub repo; open a link and **Save a copy to Drive** before editing.

Step	What to do
1	Open the Colab link from a demo lab or exercise lab
2	**File → Save a copy in Drive** (or download `.ipynb` and upload to Canvas)

Step	What to do
3	Run cells top to bottom; fix errors before submitting
4	Submit the completed `.ipynb` (and any required PDF) in Canvas

Colab home: colab.research.google.com

Colab tutorial (start here): Google Colab overview — running cells, saving a copy to Drive, editing notebooks.

Module 0 lab walks through vectors, logic, and simulation — do that before Module 1 if Colab is new to you.

Optional: R on your own computer

You do not need a local install for this course. If you prefer RStudio or Jupyter on your machine:

- **R:** CRAN · Windows/Mac/Linux installers
- **RStudio (Posit):** RStudio Desktop — install R first
- **R in Jupyter:** IRkernel setup

R is also installed on campus computers (ROSE may have free used laptops if you need hardware).

Quarto and reproducible reports

Projects and some readings use **Quarto** (`.qmd`) — text + code that renders to HTML or PDF. Optional if you want reproducible write-ups beyond plain notebooks:

- Quarto documentation
- Posit cheatsheets — R, `{tidyverse}`, `{ggplot2}`, and more

Render locally: `quarto render your-file.qmd` (requires Quarto and R).

Textbook (optional — reading only)

Introductory Statistics for the Life and Biomedical Sciences by Julie Vu and David Harrington — free PDF at OpenIntro or Leanpub. Module reads on this site are the primary path; use the textbook for alternate explanations of concepts.

Note: The textbook ships with companion R labs written in **base R**. This course uses **Google Colab**, **`{tidyverse}`**, and **`{ggplot2}`** in our own lab notebooks — follow those for R practice, not the OpenIntro lab sets.

Practice quiz banks

Open-ended practice with full answer keys on GitHub (Canvas quizzes may use a random autograded subset):

Module	Quiz A	Quiz B	Synthesis
0	Start Here Check	Question Ideas	—
1	Def	Concept	Synthesis
2	Def	Concept	Synthesis
3	Def	Concept	Synthesis
4	Def	Concept	Synthesis
5	Def	Concept	Synthesis

Optional image-based graph interpretation practice: Graph Interpretation Pack.

All quizzes are **open notes** — use your reference sheets and the methods map.

Supplemental by module

Optional **textbook chapters** (concepts) plus **videos and tutorials** (Colab + `{tidyverse}` friendly). Same links appear in each module's **Supplemental pack** row on the course path.

Module	Textbook (Vu & Harrington)	Videos & tutorials
0	Preface / book home	Colab tutorial · Posit: Getting Started with R · Navarro: R for beginners
1	Ch 2 Probability · Ch 3 Distributions (§3.2 binomial)	Khan: binomial distribution · Khan: visualizing binomial
2	Ch 3.3 Normal · Ch 4 Foundations (§4.1–4.2)	Khan: confidence intervals · Khan: CI simulation · Mine Çetinkaya-Rundel (search: confidence intervals)
3	Ch 1 Intro to data (§1.4–1.7)	R4DS: Data visualization · Posit: ggplot2 video
4	Ch 5 Inference for numerical data (§5.3) · Ch 8 Inference for categorical data (§8.1–8.2)	R4DS: Data transformation · Andrew Heiss (workflow, comparing groups)
5	Ch 6 Simple linear regression	Khan: regression line · Penn State: prediction intervals · Regression shuffle applet

Site-wide companions

Resource	Link	Best for
R for Data Science	r4ds.hadley.nz	`{tidyverse}` + `{ggplot2}` (Modules 3–5)
Posit cheatsheets	posit.co/resources/cheatsheets	Building your R reference
Posit videos	posit.co/resources/videos	Beginner R workflow
Mine Çetinkaya-Rundel	youtube.com/@minecr	Statistics-first explanations

Instructor-facing curation notes: `docs/best-practices/colab-r-companion-resources.md` in the course repo.

Getting help

- **Email:** rho3@humboldt.edu (Cal Poly Humboldt account; allow one business day during the session)
- **Office hours:** By appointment — see syllabus
- **Canvas:** Announcements, grades, and module discussions
- **Technology:** Plan ahead for Colab/Canvas outages; keep local copies of notebooks

Campus support

- **Computer labs:** R installed on campus machines
- **ROSE:** Reusable Office Supply Exchange — free used computers when available

- **SDRC:** Student Disability Resource Center — accommodations (email instructor your letter of accommodation)
